Secret-Learn — 六层架构 · 隐私计算数据流

sklearn-compatible API → privacy mode → JAX / SecretFlow compute → secure devices

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

APPLICATION · 多方隐私数据

Healthcare / Finance
cross-party modeling

IoT / Research
distributed learning

SKLEARN API · 统一算法入口

secretlearn.*
fit · predict · transform

algorithm_classifier.py
classify fit signature / epochs

PRIVACY MODES · execution placement

FEDERATED LEARNING
PYU local compute
HEU secure aggregation

SECRET SHARING
SPU MPC · ABY3 / CHEETAH

SPLIT LEARNING
split model · encrypted activations

ACCELERATION & COMPUTE

JAX Acceleration
JIT · vmap · xlearn

SecretFlow Runtime
party orchestration

SECURE DEVICES

PYU
party-local plaintext

HEU
homomorphic aggregation

SPU
secure MPC

TEE
optional enclave

INFRASTRUCTURE & HARDWARE

SecretFlow
distributed secure runtime

JAX / XLA
CPU · GPU · TPU

sklearn
API compatibility

同一 sklearn 调用由 FL / SS / SL 决定数据驻留与执行设备；JAX 加速可公开的本地计算。